

Problem 1: Build these rows with heights 1, 2, 3, and 4. Look from both ends. Count visible towers; a taller tower hides any shorter tower behind it.

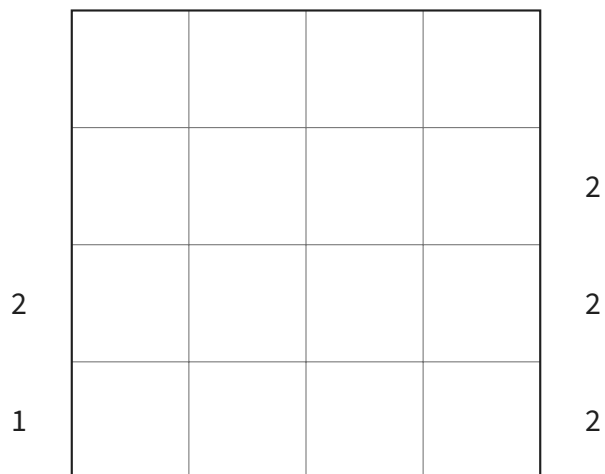
Order, left to right	Seen left	Seen right
1 4 2 3		
2 1 4 3		
4 1 2 3		
4 2 1 3		

Problem 2: Build a city using heights 1-4 once in every row and once in every column. Each outside number counts towers seen from that end. Solve the city below and record each tower height.

	4				
4					
					2
2					2
1					2

One tower I placed and the clue I used:

Problem 3: Remove the top clue 4 and the left clue 4. Rebuild using only the five remaining clues below. Keep a record of any different city you find.



Problem 4: Use four towers to test bottom rows before filling the city. Record every bottom row that shows one tower from the left and two from the right. Keep both possibilities available.

Possible bottom rows, left to right:

Problem 5: Now build rows showing two towers from each end. Record all the different rows you find. These are candidates for row 3, counting rows from the top.

Possible row 3 orders:

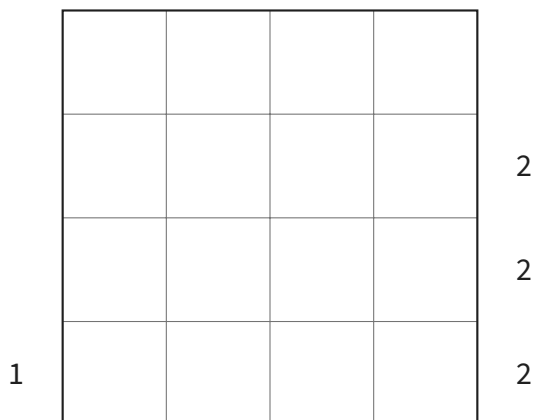
Problem 6: Pair each bottom row from Problem 4 with each row 3 from Problem 5. Reject a pair if a column repeats a height. Record the pairs that survive. For each pair, try every row 2 that avoids column repeats and shows two towers from the right. Write all the row 2 orders that work, or write “none.”

Bottom row	Row 3	Row 2 orders that work

Problem 7: For each case that survives Problem 6, fill the top row with the heights missing from its columns. Record the completed city. Check all five clues again.

				2
2				2
1				2

Problem 8: Start with the five-clue city from Problem 7. Cover the left clue on row 3. Try to make a different city that keeps the other four clues. Record it, then restore the covered clue.



Problem 9: Check the alternatives below one at a time. L3 means the left clue on row 3; R2 means the right clue on row 2. In each city, a slash begins the next row, from top to bottom. Check every row and column and the four retained clues. Record the actual view at the covered clue. Restore it before the next check.

Cover	Alternative city	Actual view
L3	1234 / 3412 / 2341 / 4123	
L4	1234 / 4123 / 3412 / 2341	
R2	1324 / 2431 / 3142 / 4213	
R3	1324 / 3142 / 2431 / 4213	
R4	1234 / 2143 / 3412 / 4321	

Problem 10: Remove every old clue. Use only left clues 4, 3, and 2 on rows 1, 2, and 3. Build a city with these clues and record its tower heights.

4				
3				
2				

Problem 11: List the top rows that work with its clue. Then list the row 2 orders that fit its clue and avoid repeats under the top row. Continue with row 3. Fill row 4 with each column's missing height.

Row	Orders that fit all earlier rows and this row's clue
1	
2	
3	
4	

Problem 12: Compare this city with your five-clue city. Record how many clues each puzzle used and whether the completed cities agree.

My comparison: